

Trainings ▾

General Information

Oil Property Guidelines for Centinel™

The Centinel™ system is designed to maintain oil quality in an acceptable range. This is accomplished by blending a portion of the oil from the oil pan with fuel to be burned. This oil is replaced by fresh oil from an onboard oil tank or by oil added manually to the engine oil sump during the daily preventative maintenance process. The result of this practice is that the oil quality and the oil contaminant levels will equilibrate to levels that are within the range for proper operation.

The Guidelines for Oil Condemnation from the Cummins Engine Oil Recommendations, Bulletin 3810340 (</qs3/pubsys2/xml/en/bulletin/3810340.html>), generally apply, but **must** be modified and expanded for the Centinel™ operation. Refer to Centinel™ Oil Condemnation Limits in the Cummins Engine Oil Recommendations, Bulletin 3810340 (</qs3/pubsys2/xml/en/bulletin/3810340.html>). Used oil analysis can help monitor engine condition when viewed with the awareness of the Centinel™ system and the guidelines shown in the table below. This document is focused on guidelines for the Centinel™ system.

Oil Analysis

Used oil analysis services typically present the data in a convenient graphic form and/or a tabular format and apply statistical techniques to the trends. In this way, alerts to check make-up oil levels and system functions can be given. The normal alert limits can be tightened to take into account the used oil removal and fresh oil dilution of the Centinel™ system.

Wear metal (iron, lead, copper, and aluminum) levels can still provide an indication of engine health when properly trended. As a guideline, a deviation in the trend line for wear metals by more than 50 percent of the mean value for the past two samples **must** alert the operator to check for potential problems. This level of wear metal increase, or exceeding the guidelines in the table, would be sufficient to justify investigation of the engine, Centinel™ system, maintenance practice, or operating procedure in order to identify the reason for the contaminant or oil property change. If the Centinel™ fault codes confirm a system problem or if the used oil values continue to deviate from the norms, an oil change is justified.

For more detailed discussion, refer to Cummins Engine Oil Recommendations, Bulletin 3810340 (</qs3/pubsys2/xml/en/bulletin/3810340.html>).

Recommended Oil Sampling Technique

- Warm the engine to proper operating temperature.
- While the engine is running at idle speed, take an oil sample after the lubricating oil filter.

With a properly functioning Centinel™ system, the values for wear metals and contaminants in the used oil will equilibrate to a steady level. This contaminant level **must** stay within a narrow band so long as the Centinel™ system is functioning properly. Changes in the duty cycle will change the oil burn and make-up rates and new equilibrium levels can be reached. If the oil make-up tank becomes empty or the system fails to function for some other reason or an engine wear problem develops, the levels of wear metals and contaminants will rise. For this reason, used oil data **must** be carefully checked and charted.

The engine lubricating oil **must** be sampled for diagnostic analysis every 250 hours for high-horsepower engines. It is strongly recommended that engine lubricating oil be sampled for diagnostic analysis every 80,470 kilometers [50,000 miles] for heavy-duty engines.